

fixes-python

Python merge-service fixes — search-flow audit 2026-06-15

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Anti-bluff (§11.4): every fix RED-first (test reproduces the defect on the pre-fix code), permanent §11.4.135 regression guard added, GREEN proven by pytest + LIVE curl against the running stack.

BUG-1 — provider selection (CRITICAL)

Root cause: only “search all enabled providers” existed; no wire field, no orchestrator param, no plugin filter for single/subset. Single/selected UI modes silently behaved as “all”.

Fix: - api/routes.py — SearchRequest gains

trackers: list[str] | None = None; POST /search and POST /search/sync pass request.trackers down. - merge_service/search.py — new helper _select_trackers(self, tracker_filter) (calls _get_enabled_trackers(), then if filter non-empty keeps only trackers whose .name is in the case-insensitive filter set). start_search(... trackers=None) persists the filter on SearchMetadata.tracker_filter; _run_search reads it back via _select_trackers; search(... trackers=None) threads it through. Empty/None == all (back-compat).

RED tests: tests/unit/merge_service/test_provider_selection.py - test_single_provider_filters_fanout — trackers=[“rutracker”] → dispatched set == {“rutracker”}, metadata.trackers_searched == [“rutracker”]. - test_subset_provider_filters_fanout — two-element subset. - test_none_filter_fans_out_to_all / test_empty_filter_fans_out_to_all — back-compat. - test_filter_is_case_insensitive — “RuTracker” → {“rutracker”}. - test_unknown_name_yields_empty_fanout_gracefully — unknown name

→ zero dispatch, no crash. -

test_start_search_seeds_filtered_trackers — async POST path seeds only the subset. Asserts on actual dispatched tracker set + metadata.trackers_searched, not status codes.

BUG-6 — integer-size crash in dedup fallback

Root cause: deduplicator._parse_size_to_bytes passed a raw int into re.match (no str()), raising TypeError; unguarded _update_best_quality→merge_results aborted the whole merge → zero results.

Fix: merge_service/deduplicator.py _parse_size_to_bytes coerces str(size_str) after the falsy guard (mirrors the api.routes / _parse_size fix). _fallback_quality signature widened to object.

RED tests: tests/unit/merge_service/test_dedup_integer_size.py - test_parse_size_to_bytes_accepts_integer / _accepts_negative_sentinel. - test_fallback_quality_survives_integer_size. - test_merge_survives_integer_size — forces the api.routes ImportError branch, int+(-1) sizes; asserts non-empty merge.

BUG-7 — all-tracker failure indistinguishable from empty

Root cause: every-provider-fail completed as status=“completed”, total_results=0 → dashboard “No results found.”, indistinguishable from genuinely empty.

Fix: merge_service/search.py _run_search — when merged==0 AND total_results==0 AND every searched tracker stat is error/timeout, set status = “all_trackers_errored” (else “completed”). Per-tracker errors already accumulate in metadata.errors.

RED tests: tests/unit/merge_service/test_all_trackers_error.py - test_all_trackers_error_is_distinguishable — all raise → status != “completed”, len(errors)==3. - test_genuinely_empty_stays_completed — no errors + 0 results stays “completed”, errors==[].

VERIFY

pytest GREEN (the 3 new regression files)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/
test_provider_selection.py \
    tests/unit/merge_service/test_dedup_integer_size.py \
    tests/unit/merge_service/test_all_trackers_error.py -v --
import-mode=importlib
...
===== 13 passed in 0.78s
=====
```

RED proof (pre-fix, same files): 12 failed / 1 passed — provider tests reject the trackers= kwarg, BUG-6 raises TypeError: expected string or bytes-like object, got 'int', BUG-7 reports status=="completed".

Full merge_service unit suite — no regression

```
$ .venv/bin/python -m pytest tests/unit/merge_service/ -q --
import-mode=importlib
818 passed, 3 warnings in 84.99s
```

ruff: All checks passed! on the 3 source + 3 test files. (Pre-existing, unrelated: tests/unit/test_plugin_yts_deep.py fails 45 on a pristine stashed tree — collection error in the YTS plugin test, not touched by these fixes.)

LIVE — running stack (podman VM, merge service :7187)

After editing: cleared __pycache__, podman restart qbittorrent-proxy, verified served code matches committed edits (_select_trackers, all_trackers_errored, size_str = str(size_str) all present in /config/download-proxy/...).

```
=== FILTERED trackers_searched (rutor only) ===
status: completed | total: 3 | trackers_searched: ['rutor']
```

```
=== NO-FILTER (all enabled providers) ===
status: completed | total: 1182 | n_trackers: 29 |
trackers_searched:
['rutracker', 'kinozal', 'nnmclub', 'iptorrents', 'jackett', 'academictr
'anilibra', 'bitsearch', 'gamestorrents', 'glotorrents', 'kickass', 'lir
'linuxtracker', 'megapeer', 'nyaa', 'piratebay', 'pirateiro', 'rockbox',
'snowfl', 'tokyotoshokan', 'torlock', 'torrentdownload', 'torrentgalaxy
'torrentkitty', 'torrentproject', 'torrentscsv', 'yourbittorrent', 'yts
```

Filtered search hits ONLY rutor (3 results); no-filter hits all 29 providers (1182 results) — provider selection proven end-to-end through the real HTTP path.

Files changed: download-proxy/src/api/routes.py, download-proxy/src/merge_service/search.py, download-proxy/src/merge_service/deduplicator.py. Tests added: tests/unit/merge_service/test_provider_selection.py, test_dedup_integer_size.py, test_all_trackers_error.py. NOT committed.